

Rassul Zeinulla

Robotics AI & Mechatronics Engineer

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📁 Portfolio | 🌐 Rassul Zeinulla | 🌐 rassulz

Research Interests

Reinforcement learning for robot control; vision–language–action (VLA) models; computer vision for robotic perception; robotic manipulators; legged robots.

Education

Nazarbayev University, *Master's in Robotics* Aug 2026 – July 2028

Kazakh-British Technical University, *B.S. in Automation and Control* Sept 2022 – May 2026

ABET-accredited program, German Anabin database.

Relevant coursework: Linear and Nonlinear Control Systems, Autonomous Mobile Robots, Industrial Robot Operations, Machine Learning, Robotics Estimation and Learning, and Computer Vision.

Technical Skills

Programming Languages: C++, Python, MATLAB, IEC 61131-3 (ST, LD, FBD)

Robotics & Simulation: ROS 1 Noetic, ROS 2 Humble, RViz2, MoveIt 2, Autodesk Fusion 360, HD Hyundai Robotics HRSpace, NVIDIA Isaac Sim

Computer Vision & Machine Learning: YOLO, OpenCV, PyTorch, AI agents, Transformers, scikit-learn

Automation & Control: Siemens TIA Portal, Schneider EcoStruxure Control Expert, Honeywell UniSim Design

Electronics and Hardware Skills: Raspberry Pi 4/5, ESP32, NVIDIA Jetson, Intel RealSense D435i, Schneider Electric Modicon M340 and M241 controllers, SIMATIC S7-1200/S7-300 PLCs, KUKA KRC4 compact controller, Kawasaki Astorino

Software & Tools: Git, GitHub, Linux, Docker

Accepted Conference Paper

R. Zeinulla, A. Abdimalik, A. Rakhmetkali, C. Masimba, and T. Samigulin, “Distributed Control Framework of Multirobot System Using Automation and Control Applications,” accepted to the *2026 IEEE 20th International Conference on Control and Automation (ICCA)*, Almaty, Kazakhstan, June 2026.

This paper describes the Multirobot Control System for Automated Part Painting project listed below.

Experience

Machine Learning Trainee – Summer Program, *Yessenov Data Lab* June 2026 – July 2026

- **Semantic search:** Built an offline image-search application over a 3,000-image Flickr8k subset using CLIP embeddings and cosine-similarity retrieval, with a FastAPI backend and React interface.
- Built a local Russian-to-English translator using MarianMT and Streamlit.

Robotics Intern (Short-Term), *Arcobo* March 2026 – April 2026

- **Kawasaki BA006L:** Deployed and operated the industrial robot; programmed trajectory-following and pick-and-place tasks on the Kawasaki Astorino platform.
- Validated robot motion workflows with SimBox HIL simulation before execution.

Robotics Laboratory Assistant, *Kazakh-British Technical University* Jan 2025 – Jan 2026

- Co-authored a practical laboratory manual for NVIDIA Isaac Sim.
- **Self-balancing platform:** Designed and implemented a system with two BLDC motors, an Arduino Uno, MPU-6050 feedback, and a PID controller modeled in MATLAB/Simulink.

- Integrated computer-vision models into laboratory projects.

Automation & Robotics Engineer Intern, Hyundai Trans Kazakhstan

June 2025 – Aug 2025

- **SIMATIC S7-300:** Programmed and tested ladder logic for a conveyor-control training system in TIA Portal, including start/stop, direction, emergency-stop, timer sequences, and sensor I/O.
- Assembled and wired its control cabinet, installed and calibrated inductive proximity sensors, and built XP-Builder HMI screens for controls, alarms, and real-time system-state visualization.
- Participated in Hyundai Accent body-model integration and trajectory-path optimization for robotic spot-welding cells; studied AGV intralogistics and PLC communication protocols.

Selected Projects

Kawasaki ROS 2 Safety-Zone Monitoring

 [GitHub repository](#) 

- **Safety response:** Developed and validated a virtual industrial-safety workflow in NVIDIA Isaac Sim 5.0.0 and ROS 2 Humble: Ultralytics YOLO monitored a simulated RealSense D455 feed and triggered stop/resume behavior for a Kawasaki manipulator controlled through inverse kinematics.
- Tools used: Python, ROS 2 Humble, NVIDIA Isaac Sim 5.0.0, OpenCV, Ultralytics YOLO, virtual Intel RealSense D455.

Industrial Machine Learning with PLC Integration

 [GitHub repository](#) 

- **100,000 rows:** Co-developed and evaluated regression models on iron-ore flotation data to predict silica concentration and compared linear and random-forest approaches.
- The team integrated the prediction workflow with a SIMATIC S7-1200 PLC through MATLAB and OPC UA for HMI visualization.
- Tools used: MATLAB, Python, pandas, NumPy, scikit-learn, HMI, SIMATIC S7-1200, OPC UA.

Multirobot Control System for Automated Part Painting

 [GitHub repository](#) 

- **Multi-robot cell:** Co-developed a laboratory-validated automated painting system coordinating two RoArm-M2-S manipulators and a KUKA KR10 with an S7-1500-controlled conveyor.
- The system integrates MATLAB/Simulink supervisory control with PLC sequencing through OPC UA and a WinCC HMI for real-time monitoring, safety interlocks, and coordinated operation.
- Tools used: KUKA KR10 R1100-2, two RoArm-M2-S manipulators, BLDC conveyor motor, Siemens SIMATIC S7-1500, TIA Portal V19, WinCC HMI, MATLAB, Simulink, OPC UA.

Awards & Leadership

Second Place, University Course Project: Ranked 2nd among 40+ teams in the university's fourth-year Automation and Control cohort.

Robo-Football Hackathon Mentor & Judge: Mentored student teams and served on the judging panel at ITFEST 2024 and ITFEST 2025.

KBTU Robotics Outreach: Represented the Robotics and Automation program at Open Day and EDUFEST, advising prospective students and demonstrating student projects.

Portfolio

Feel free to view my detailed projects and demonstrations in my  [Portfolio](#) .